SMARTBRIDGE
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SKILL WALLET
A SmartBridge Product

Project Initialization and Planning Phase

Date	26 June 2026
Team ID	
Project Title	Rising Waters – A Flood Prediction
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposed solution aims to provide an intelligent flood prediction system using Machine Learning techniques. The system analyzes historical weather and environmental data to predict flood risk accurately. It helps government authorities, disaster management teams, and citizens receive early flood warnings, enabling timely preventive actions and reducing the impact of flood disasters.

Project Overview	
Objective	Develop an accurate Machine Learning-based flood prediction system that analyzes weather and environmental parameters to provide early flood risk predictions and support disaster preparedness.
Scope	The project focuses on collecting historical flood-related datasets, preprocessing data, training multiple Machine Learning models, selecting the best-performing model, and integrating it into a Flask web application for real-time flood risk prediction.
Problem Statement	
Description	Floods cause severe damage to human life, agriculture, infrastructure, and the environment. Existing warning systems often provide delayed or less accurate predictions, making it difficult for authorities and citizens to take timely preventive measures.
Impact	An accurate flood prediction system enables early warning, improves disaster preparedness, minimizes economic losses, protects lives and property, and assists government agencies in effective emergency response planning.
Proposed Solution	
Approach	Develop a Machine Learning-based flood prediction system that preprocesses historical weather data, trains multiple predictive models (Decision Tree, Random Forest, KNN, and XGBoost), selects the most accurate model, and deploys it through a Flask web application

	for user-friendly flood risk prediction.
Key Features	1. Predict flood risk using Machine Learning algorithms. 2. Analyse weather and environmental parameters. 3. Compare multiple ML models for best accuracy. 4. Display prediction results through a Flask web interface. 5. Store prediction history for future reference. 6. Support disaster preparedness through early flood warnings.

Resource Requirements

Resource Type	**Description**	**Specification/Allocation**
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Intel Core i3 (or higher)
Memory	RAM specifications	Min 4GB RAM
Storage	Disk space for data, models, applications and logs	Min 2 GB Free SSD Space
Software		
Frameworks	Web application framework	Flask
Libraries	Additional libraries	scikit-learn, pandas, numpy, matplotlib, seaborn
Development Environment	IDE	Jupyter Notebook, Pycharm
Data		
Data	Source, size, format	Kaggle dataset – Flood Risk In India Dataset